Tutorial 9

Basis, dimension and subspaces associated with a matrix

1. Suppose that $V = \mathbb{R}^m$. For each of the following statements say whether it is true or false, and give a reason for your answer. You may quote without proof any theorem from the notes. (This should also go quickly!)
   (a) Every basis for V has m vectors.
   (b) Every set of k vectors of V with $k < m$ is linearly independent.
   (c) Every set of k vectors of V with $k > m$ is linearly dependent.
   (d) Every set of k vectors of V with $k < m$ fails to span V.
   (e) Every set of k vectors of V with $k > m$ spans V.

2. (a) Let $C = \{x^3, x^2 + 1, x\}$.
   i. Is C a basis for P_3? Explain briefly why or why not.
   ii. If C is not a basis for P_3, can it be enlarged to a basis by adding appropriate vectors? If it can be enlarged to a basis, do this; otherwise explain why it cannot be done.

   (b) Let $C = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$.
   i. Is C a basis for $\mathbb{R}^4$? Explain briefly why or why not.
   ii. If C is not a basis for $\mathbb{R}^4$, can it be enlarged to a basis by adding appropriate vectors? If it can be enlarged to a basis, do this; otherwise explain why it cannot be.

   (c) Let $C = \left\{ \begin{pmatrix} 0 & 0 \\ 0 & 7 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 6 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 6 & 7 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 15 & 6 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \right\}$.
   i. Is C a basis for $M_{2\times 2}(\mathbb{R})$? Explain briefly why or why not.
   ii. If C is not a basis for $M_{2\times 2}(\mathbb{R})$, can it be reduced to a basis by omitting appropriate vectors? If it can be reduced to a basis, do this; otherwise explain why it cannot be.

3. The trace of an $n \times n$ matrix $A = (a_{ij})$, denoted by $\text{tr}(A)$, is the sum of the entries on the diagonal of A, i.e. .

$$\text{tr}(A) = \sum_{i=1}^{n} a_{ii}.$$

   (a) Show that $S = \{A \in M_{n\times n}(\mathbb{R}) : \text{tr}(A) = 0\}$ is a subspace of $M_{n\times n}(\mathbb{R})$
   (b) Find a basis for S.
   (c) What does $\dim(S)$ equal?

4. Let $A = \begin{pmatrix} 1 & 2 & 3 & 2 & 1 \\ 3 & 7 & 10 & 5 & 5 \\ 1 & 3 & 4 & 0 & 3 \\ 6 & 15 & 21 & 7 & 12 \end{pmatrix}$.

   We can row reduce A to U, where $U = \begin{pmatrix} 1 & 2 & 3 & 2 & 1 \\ 0 & 1 & 1 & -1 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

   (**You do not need to check this!**) Which of the following statements are true? (You should be able to answer this question very quickly; **use your notes.**)
   (a) $NS(A) = NS(U)$.
   (b) $RS(A) = RS(U)$.
   (c) $CS(A) = CS(U)$.
   (d) $\text{rank}(A) = \text{rank}\ (U)$
   (e) nullity (A)=nullity (A^T)

5. Let A be the matrix in Question 4. Answer the following questions, using where appropriate, the fact that A can be row-reduced to the matrix U in Question 4.
   (a) Write down a basis for, and the dimension of, $NS(A)$.
   (b) Write down a basis for, and the dimension of, $RS(A)$.
   (c) Write down a basis for, and the dimension of, $CS(A)$.
   (d) What is the nullity of A?
   (e) What is the rank of A?
   (f) What is the nullity of A^T?

6. (a) Suppose A is a 11×46 matrix.

   (i) What is the maximum possible value for the rank of A?

   (ii) What is the minimum possible value for the rank of A?

   (iii) What is the maximum possible value for the nullity of A?

   (iv) What is the minimum possible value for the nullity of A?

   (b) Answer the same questions as (a) for a 46×11 matrix A.

7. Let V be a vector space and B be a basis for V.

   (a) Prove that if S is a linearly independent subset of V which contains B, then $S = B$. (This is sometimes expressed by saying a basis is a *maximal linearly independent* subset.)

   (b) Prove that if S is a subset of V which spans V and is contained in B, then $S = B$. (This is sometimes expressed by saying a basis is a *minimal spanning* subset.)

   (c) Prove that a maximal linearly independent subset of V is a basis for V.

   (d) Prove that a minimal spanning subset of V is a basis for V.